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Assignment 1
Course: CSE 242
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Part 1

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

Part 1 Q1

Compute AB, BA, $(A+B)^T$, AB^T, $\text{trace}(A)$, $\text{trace}(AB)$.

Solution:

$$\begin{aligned} AB_1 &= 1A_1 + 0A_2 \\ &= A_1 \\ &= \begin{bmatrix} 3 \\ 2 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} AB_2 &= 2A_1 - 1A_2 \\ &= \begin{bmatrix} 6 \\ 4 \end{bmatrix} - \begin{bmatrix} 4 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 3 \end{bmatrix} \end{aligned}$$

$$\boldsymbol{AB} = [\boldsymbol{AB_1} \quad \boldsymbol{AB_2}] = \begin{bmatrix} \mathbf{3} & \mathbf{2} \\ \mathbf{2} & \mathbf{3} \end{bmatrix}$$

OK now you can see that I know how to do matrix multiplication so I will do the rest in an easier to write way.

$$\begin{bmatrix} 3\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 2 \\ -1 \end{bmatrix} & 4\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 2 \\ -1 \end{bmatrix} \end{bmatrix}$$

$$\begin{bmatrix} \begin{bmatrix} 3 \\ 0 \end{bmatrix} + \begin{bmatrix} 4 \\ -2 \end{bmatrix} & \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \end{bmatrix} \end{bmatrix}$$

$$\begin{bmatrix} \begin{bmatrix} 7 \\ -2 \end{bmatrix} & \begin{bmatrix} 6 \\ -1 \end{bmatrix} \end{bmatrix}$$

$$\boldsymbol{BA} = \begin{bmatrix} \mathbf{7} & \mathbf{6} \\ \mathbf{-2} & \mathbf{-1} \end{bmatrix}$$

$$B^T = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$$

$$AB^T = \begin{bmatrix} [3 \;\; 4] \cdot [1 \;\; 2] & [3 \;\; 4] \cdot [0 \;\; -1] \\ [2 \;\; 1] \cdot [1 \;\; 2] & [2 \;\; 1] \cdot [0 \;\; -1] \end{bmatrix}$$

$$= \begin{bmatrix} 3+8 & -4 \\ 2+2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{11} & \mathbf{-4} \\ \mathbf{4} & \mathbf{-1} \end{bmatrix}$$

$$A + B = \begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix}$$

$$(\boldsymbol{A} + \boldsymbol{B})^{\boldsymbol{T}} = \begin{bmatrix} \mathbf{4} & \mathbf{2} \\ \mathbf{6} & \mathbf{0} \end{bmatrix}$$

The trace wuz sum of the diagonals right? $\mathbf{trace}(\boldsymbol{A}) = \mathbf{4}$, $\mathbf{trace}(\boldsymbol{AB}) = \mathbf{6}$

Part 1 Q2

For the following matrix:

$$M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Calculate the eigenstuff, then verify.

I admittedly looked up how to solve for eigenvectors:

The steps seem to be:
1. Solve $\det(M - \lambda I) = 0$ to get eigenvalues
   - I remember there was a nice way to get the determinant of a 2x2 matrix. Something like $ad - bc$. I'll verify this online.
2. Solve $(M - \lambda I)v = 0$ to get the eigenvectors

Solution:

$$\begin{aligned}
\det(M - \lambda I) &= \det\left(\begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix}\right) \\
&= (2-\lambda)^2 - 1 \\
&= \lambda^2 - 4\lambda + 3 \\
\lambda &= \frac{4 \pm \sqrt{(-4)^2 - 4(1)(3)}}{2(1)} \\
&= \frac{4 \pm \sqrt{16 - 12}}{2} \\
&= \frac{4 \pm 2}{2} \\
\boldsymbol{\lambda} &= \mathbf{2 \pm 1} = \{\mathbf{1}, \mathbf{3}\}
\end{aligned}$$

Case 1:

$$\begin{aligned}
\begin{bmatrix} 2-1 & -1 \\ -1 & 2-1 \end{bmatrix} v_1 &= 0 \\
\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} v_1 &= 0 \\
v_{11} - v_{12} &= 0 \\
v_{11} &= v_{12} \\
\boldsymbol{v}_1 &= \begin{bmatrix} \mathbf{1} \\ \mathbf{1} \end{bmatrix}
\end{aligned}$$

Case 2:

$$\begin{aligned}
\begin{bmatrix} 2-3 & -1 \\ -1 & 2-3 \end{bmatrix} v_2 &= 0 \\
\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} v_2 &= 0 \\
-v_{21} - v_{22} &= 0 \\
-v_{21} &= v_{22} \\
\boldsymbol{v}_2 &= \begin{bmatrix} \mathbf{-1} \\ \mathbf{1} \end{bmatrix}
\end{aligned}$$

Part 1 Q3

Prove that $\forall AB, \text{trace}(AB) = \text{trace}(BA)$

Solution:

A and B must be square matrices by premise. Let $AB = C$ and $BA = C'$. Generally, matrix multiplication for an arbitrary A and B of size n^2 is

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

We recall the definition of a trace of a matrix A of size n^2:

$$\text{trace}(A) = \sum_{i=1}^{n} A_{ii}$$

Unfold the definitions of trace and matrix multiplication in our goal.

$$\vdash \sum_{i=1}^{n} \sum_{k=1}^{n} A_{ik} B_{ki} = \sum_{i=1}^{n} \sum_{k=1}^{n} B_{ik} A_{ki}$$

Rewrite by add commutativity, swapping the order of the sums on the RHS:

$$\vdash \sum_{i=1}^{n} \sum_{k=1}^{n} A_{ik} B_{ki} = \sum_{k=1}^{n} \sum_{i=1}^{n} B_{ik} A_{ki}$$

We recognize that i and k are bound and are local to their respective $\sum$s.

$$\vdash \boxed{\sum_{\boxed{i=1}}^{n} \boxed{\sum_{\boxed{k=1}}^{n} A_{ik}B_{ki}}} = \boxed{\sum_{\boxed{k=1}}^{n} \boxed{\sum_{\boxed{i=1}}^{n} B_{ik}A_{ki}}}$$

Therefore their names are relevant only within their boxes. Our ultimate goal is to show α-equivalence and so I will:

- On the left, rewrite $i \to \star, k \to \diamond$
- On the right, rewrite $i \to \diamond, k \to \star$

$$\vdash \boxed{\sum_{\boxed{\star=1}}^{n} \boxed{\sum_{\boxed{\diamond=1}}^{n} A_{\star\diamond}B_{\diamond\star}}} = \boxed{\sum_{\boxed{\star=1}}^{n} \boxed{\sum_{\boxed{\diamond=1}}^{n} B_{\diamond\star}A_{\star\diamond}}}$$

Scalar multiplication has commutativity, and so rewrite $B_{\diamond\star}A_{\star\diamond} \to A_{\star\diamond}B_{\diamond\star}$

$$\vdash \sum_{\star=1}^{n}\sum_{\diamond=1}^{n} A_{\star\diamond}B_{\diamond\star} = \sum_{\star=1}^{n}\sum_{\diamond=1}^{n} A_{\star\diamond}B_{\diamond\star}$$

Reflexivity. QED.

Part 1 Q4

Compute the inverse of

$$C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 4 & 7 & 1 & 0 \\ 2 & 6 & 0 & 1 \end{bmatrix} \qquad \frac{r_2}{2} \to \begin{bmatrix} 4 & 7 & 1 & 0 \\ \frac{2}{2} & \frac{6}{2} & \frac{0}{2} & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} 4 & 7 & 1 & 0 \\ 1 & 3 & 0 & \frac{1}{2} \end{bmatrix} \quad r_1 - 4r_2 \to \begin{bmatrix} 4-4*1 & 7-4*3 & 1-4*0 & 0-4*\frac{1}{2} \\ 1 & 3 & 0 & \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 7-12 & 1 & -2 \\ 1 & 3 & 0 & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} 0 & -5 & 1 & -2 \\ 1 & 3 & 0 & \frac{1}{2} \end{bmatrix} \qquad \frac{r_1}{-5} \to \begin{bmatrix} 0 & \frac{-5}{-5} & \frac{1}{-5} & \frac{-2}{-5} \\ 1 & 3 & 0 & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -\frac{1}{5} & \frac{2}{5} \\ 1 & 3 & 0 & \frac{1}{2} \end{bmatrix} \quad r_2 - 3r_1 \to \begin{bmatrix} 0 & 1 & -\frac{1}{5} & \frac{2}{5} \\ 1 & 0 & \frac{3}{5} & \frac{1}{2} - \frac{6}{5} = \frac{5}{10} - \frac{12}{10} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -\frac{1}{5} & \frac{2}{5} \\ 1 & 0 & \frac{3}{5} & -\frac{7}{10} \end{bmatrix} \quad r_1 \leftrightarrow r_2 \to \begin{bmatrix} \mathbf{\frac{3}{5}} & \mathbf{-\frac{7}{10}} \\ \mathbf{-\frac{1}{5}} & \mathbf{\frac{2}{5}} \end{bmatrix}$$

Part 2

Part 2 Q1

Suppose a random variable X follows a discrete distribution:

$$P(X = k) = c \cdot \frac{3^k}{k!}, k = 0, 1, 2, ...$$

(a) Find the normalization constant c.

(b) Compute the expectation $E[X]$ and variance $\mathrm{Var}(X)$.

Solution:

$$1 = \sum_{k=0}^{\infty} c\frac{3^k}{k!}$$
$$1 = c\sum_{k=0}^{\infty} \frac{3^k}{k!}$$

We recall that the taylor series of $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ and so the above product must be ce^3 and $\boldsymbol{c := e^{-3}}$.

Just to restate it:

$$P(X = k) = e^{-3}\frac{3^k}{k!}$$

Expectation in this case would be:

$$\begin{aligned}
E[X] &= \sum_{k=0}^{\infty} ke^{-3}\frac{3^k}{k!} \\
&= 0e^{-3}\frac{3^0}{0!} + \sum_{k=1} ke^{-3}\frac{3^k}{k!} \\
&= \sum_{k=1}^{\infty} e^{-3}\frac{3^k}{(k-1)!} \\
&= e^{-3}\sum_{k=1}^{\infty} \frac{3^k}{(k-1)!} \\
&= e^{-3}\sum_{k=1}^{\infty} 3 * \frac{3^{k-1}}{(k-1)!} \\
&= 3e^{-3}\sum_{k=1}^{\infty} \frac{3^{k-1}}{(k-1)!} \\
&= 3e^{-3}\sum_{k=0}^{\infty} \frac{3^k}{k!} \\
&= 3e^{-3}e^3 \\
&= \mathbf{3}
\end{aligned}$$

Solution:

I have no idea how to solve this.

Part 2 Q2

Given two events A and B with $P(A) = 0.5$, $P(B) = 0.3$, and $P(A \mid B) = 0.6$, compute the following:

(a) $P(A \cap B)$,

(b) $P(B \mid A)$,

(c) $P(A \cup B)$.

Solution:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \qquad P(B|A) = \frac{P(A \cap B)}{P(A)} \qquad P(A \cup B)$$

$$0.6 = \frac{P(A \cap B)}{0.3} \qquad = \frac{\mathbf{0.18}}{\mathbf{0.5}} = \mathbf{0.36} \qquad = P(A) + P(B) - P(A \cap B)$$

$$\mathbf{0.6 * 0.3 = 0.18 = P(A \cap B)} \qquad = 0.5 + 0.3 - 0.18$$

$$= \mathbf{0.62}$$

Part 2 Q3

Suppose you have a biased coin with unknown probability of heads $\theta = P(H)$. You flip this coin 5 times and observe the sequence: H, H, T, H, T.

(a) Derive the Maximum Likelihood Estimate (MLE) for θ, showing each step from the likelihood function.

(b) Compute the Maximum a Posteriori (MAP) estimate for θ, assuming a prior distribution:

$$P(\theta) = \begin{cases} 2\theta & , 0 \leq \theta \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

Solution:

$$L(\theta) = \theta^3 * (1-\theta)^2 = \theta^5 - 2\theta^4 + \theta^3$$

$$L'(\theta) = 5\theta^4 - 8\theta^3 - 3\theta^2 = \theta^2(5\theta^2 - 8\theta + 3)$$

Ok and now we cheat a little bit because obviously $\theta = 0$ is a solution to $L'(\theta) = 0$ but that would mean the probability of heads is zero, so that's crazy. So we know that the right product must be zero:

$$0 = (5\theta^2 - 8\theta + 3)$$

And this is a quadratic equation so now we just use the quadratic equation formula. I don't wanna write it up. The first result is 1, which is also crazy, so we will use $b -$, which is $\boldsymbol{\theta = 0.6}$.

Solution:

Unsolved. I have no idea what this is.

Part 2 Q4

Consider a Gaussian distribution with parameters μ (mean) and σ^2 (variance). Given the dataset $X = \{1, 3, 5, 7\}$, derive the Maximum Likelihood Estimates (MLE) for both μ and σ^2. Show each mathematical step clearly.

Part 2 Q5

You observe samples drawn from a Gaussian distribution with unknown mean μ and known variance $\sigma^2 = 4$. Given a Gaussian prior for the mean μ:

Part 2 Q6

Prove the variance expansion identity:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Solution:

We recall that in all cases, expectation is a sum, and so the usual rules apply.

$$\begin{aligned}\text{Var}(X) &= E\big[(X - E[X])^2\big] \\ &= E\big[X^2 - 2XE[X] + E[X]^2\big]\end{aligned}$$

Expectation of a constant is a constant soooo

$$\begin{aligned}&= E[X^2] - 2E[X]E[X] + E[X]^2 \\ &= E[X^2] + E[X]^2 - 2E[X]^2 \\ &= \boldsymbol{E[X^2] - E[X]^2}\end{aligned}$$

Part 3 Q1

Given a dataset represented by a random vector X, derive the mathematical expression for the variance of the projection onto a unit vector u:

$$\text{Var}(u^T X) = u^T \Sigma u$$